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Hyperbolic and Dispersive 1D PDE in Lean 4

Weak Solutions, Shock Waves, the Lax Entropy Condition, Gradient Blow-up and Solitons

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Code. <<https://github.com/Alektronnik/M4TH>> — packages

ConservationLaws, BurgersBlowUp, KdV

License. Apache 2.0 (software), CC-BY 4.0 (text) **Verification.** Lean 4 over Mathlib. Zero axiom, zero sorry. Every displayed theorem carries a clean axiom certificate ([propext, Classical.choice, Quot.sound]).

Abstract

We report a self-contained formalisation, in the Lean 4 proof assistant on top of Mathlib, of the elementary theory of one-dimensional evolutionary partial differential equations of hyperbolic and dispersive type. Three results anchor the development. First, for the scalar conservation law $\partial_t u + \partial_x(f(u)) = 0$ with an *arbitrary* flux, prove that the distributional residual of a travelling step against any smooth compactly supported test function reduces *exactly* to the boundary integral of the Rankine–Hugoniot deficit along the moving interface; as a corollary the step is a weak solution if and only if its speed satisfies Rankine–Hugoniot. For the Burgers flux we formalise the Lax entropy pair, the entropy-dissipation identity, and the classical non-uniqueness pathology (a genuine weak solution that violates the entropy condition). Second, for the inviscid Burgers equation we prove finite-time gradient blow-up for a compressive datum, by an exact solution of the Riccati equation governing the gradient along characteristics, with a purely algebraic final contradiction. Third, for the Korteweg–de Vries equation we verify that the classical squared-hyperbolic-secant profile is an exact travelling-wave solution

and that mass and quadratic energy are conserved for compactly supported solutions. To our knowledge this is the first formalisation of weak solutions of conservation laws, of the Rankine–Hugoniot and Lax entropy conditions, of gradient blow-up for a nonlinear PDE, and of the KdV soliton, in any major proof assistant. The material is organised as three independent packages, each split into a linear series of self-contained modules designed for incremental contribution to Mathlib.

1. Introduction

The formal verification of analysis has advanced rapidly, but the theory of partial differential equations has remained almost entirely outside the reach of proof assistants. The recent formalisation of the De Giorgi–Nash–Moser theorem (2026) is, to our knowledge, the first non-trivial regularity result for PDE in Lean; its own authors note the prior view that non-trivial statements about solutions of PDE were not formalisable. That result concerns *elliptic* regularity. The complementary world — *evolutionary* equations, and in particular the hyperbolic and dispersive phenomena of shock formation, entropy selection, finite-time singularities, and solitary waves — has, as far as we have been able to determine, no counterpart in any major library.

This paper occupies that territory with elementary but foundational material. We deliberately restrict to one spatial dimension and to the most classical model problems, because our aim is not maximal generality but the establishment of a *verified vocabulary* : the definitions of weak solution, Rankine–Hugoniot speed, entropy pair, characteristic, and travelling wave, together with the theorems that make these definitions non-vacuous. Each definition is chosen to be the one a subsequent Mathlib development would want to build on, and each is accompanied by at least one concrete instance in which the corresponding phenomenon is exhibited in full.

Three theorems organise the development.

1. **Exact shock reduction** (Section 3). For the scalar conservation law with

arbitrary flux f , the distributional residual of the travelling step $\text{shockProfile } u_L \ u_R$ collapses exactly to the Rankine–Hugoniot deficit integrated along the interface. This is an equality, not an estimate, and it holds for every admissible test function; the Rankine–Hugoniot condition is recovered as the precise vanishing criterion.

2. **Gradient blow-up** (Section 5). For the inviscid Burgers equation with the

compressive ramp datum $u_0(x) = -x$, no C^2 solution survives to time 1. The proof tracks the spatial gradient along characteristics, where it solves the Riccati equation $V' = -V^2$ with $V(0) = -1$, whose exact solution $V(t) = -1/(1 - t)$ diverges at $t = 1$; the contradiction with putative gradient bound is algebraic and uses no limiting argument.

3. **The KdV soliton** (Section 6). The profile $3c \cdot \text{sech}^2((\sqrt{c}/2) \cdot \xi)$ is an exact

travelling-wave solution of $u_t + u u_x + u_{xxx} = 0$ for every wave speed $c > 0$, and mass and quadratic energy are conserved for compactly supported solutions.

All three are verified with no axioms beyond the foundations of Lean and Mathlib and with no incomplete proofs. The development is partitioned into three packages that share no code and depend only on Mathlib, so that each can be reviewed, built, and upstreamed independently.

1.1 Contributions

- The first formalisation of weak (distributional) solutions of scalar conservation laws, for arbitrary flux, with a measure-theoretic

residual over the open space–time cylinder.

- An exact reduction of the weak residual of a travelling discontinuity to the

Rankine–Hugoniot deficit, and the equivalence "weak solution $\Leftrightarrow$ Rankine–Hugoniot".

- The Lax entropy pair for the Burgers flux, the closed-form entropy–dissipation

identity, and a formal witness of non-uniqueness: an expansion step that is a weak solution but is not entropy–admissible.

- The first formalisation of finite-time gradient blow-up for a nonlinear PDE,

via an exact Riccati solution and a Lyapunov–energy uniqueness lemma for the characteristic ODE.

- The first formalisation of the KdV one-soliton and of its mass and energy

conservation laws.

1.2 Non-goals

We do not construct global weak solutions, prove existence via vanishing viscosity or front tracking, treat systems or several space dimensions, or establish uniqueness in an entropy class. These are the natural next steps; the present work supplies the definitions and the first theorems on which they would rest. Where a result is special to the Burgers flux (the entropy pair, the ramp blow-up) we mark it as such, and we keep the general-flux theory strictly separate from the Burgers instance.

2. Setting and Mathlib background

All three packages are stated over the reals and use only core Mathlib analysis: `ContDiff` and `HasCompactSupport` for smoothness and support, `deriv` and `fderiv` for differentiation, `MeasureTheory` for the space–time integral, and the ODE uniqueness and Grönwall infrastructure of `Mathlib/Analysis/ODE`. No differential-geometric machinery is required: in one space dimension the objects are ordinary functions $\mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R}$ of time and space, and the analytical content is carried by elementary calculus, the fundamental theorem of calculus in its improper form, and Fubini's theorem.

The space–time domain is the open cylinder

```
def spacetimeDomain (T : ℝ) : Set (ℝ × ℝ) := Set.Ioo 0 T ×ˢ Set.univ
```

and test functions are smooth, compactly supported, and supported inside it.

3. Weak solutions of scalar conservation laws

3.1 Test functions and the weak residual

A test function on the cylinder $(0, T) \times \mathbb{R}$ is packaged as a structure carrying its own smoothness and support proofs.

```
structure TestFunction (T : ℝ) where
  toFun : ℝ → ℝ → ℝ
  smooth : ContDiff ℝ (τ : ℕ) (fun p : ℝ × ℝ => toFun p.1 p.2)
  compactSupport : HasCompactSupport (fun p : ℝ × ℝ => toFun p.1 p.2)
  support_subset : tsupport (fun p : ℝ × ℝ => toFun p.1 p.2) ⊆ spacetimeDomain T
```

Writing `φ.dt` and `φ.dx` for the partial derivatives of the test function the weak formulation of $\partial_t u + \partial_x (f(u)) = 0$ transfers all derivatives onto ϕ :

```

noncomputable def weakIntegrand (f : ℝ → ℝ) (u : ℝ → ℝ → ℝ)
(φ : TestFunction T) (t x : ℝ) : ℝ :=
u t x * φ.dt t x + f (u t x) * φ.dx t x

noncomputable def weakResidual (f : ℝ → ℝ) (u : ℝ → ℝ → ℝ)
(φ : TestFunction T) : ℝ :=
∫ t in Ioo 0 T, ∫ x : ℝ, weakIntegrand f u φ t x

```

A function is a weak solution when it is locally integrable together with its flux and its residual vanishes against every test function (`IsWeakSolution`). Because the support of ϕ is compact and interior to the cylinder, the inner spatial integral is over a bounded interval and all integrals are finite.

3.2 The travelling step and Rankine–Hugoniot

The elementary discontinuous profile is the travelling step, together with the Rankine–Hugoniot speed condition.

```

noncomputable def shockProfile (uL uR s : ℝ) (t x : ℝ) : ℝ :=
if x < s * t then uL else uR

def RankineHugoniot (f : ℝ → ℝ) (uL uR s : ℝ) : Prop :=
s * (uL - uR) = f uL - f uR

```

3.3 The exact shock reduction

The central theorem of the package evaluates the weak residual of the step in closed form. After a Galilean change of frame that straightens the interface and an application of Fubini and the improper fundamental theorem of calculus, every contribution away from the interface cancels and only the jump survives.

```

theorem ConservationLaw.hasShockIntegralReduction (f : ℝ → ℝ) (T uL uR s :
ℝ) :
∀ φ : TestFunction T,
weakResidual f (shockProfile uL uR s) φ =
∫ t in Ioo 0 T, ((f uL - f uR) - s * (uL - uR)) * φ t (s * t)

```

The integrand is the Rankine–Hugoniot deficit $(f \ uL - f \ uR) - s(uL - uR)$

u_R) weighted by the trace of the test function along the interface $x = s$. Two corollaries follow immediately: if the speed satisfies Rankine–Hugoniot the deficit is zero and the residual vanishes, so the step is a weak solution; and, conversely, a step that is a weak solution must satisfy Rankine–Hugoniot.

```
theorem ConservationLaw.isWeakSolution_shockProfile_of_rankineHugoniot
{f : ℝ → ℝ} (h : RankineHugoniot f uL uR s) :
IsWeakSolution f T (shockProfile uL uR s)
```

We emphasise that these statements hold for an *arbitrary* flux $f : \mathbb{R} \rightarrow \mathbb{R}$; no convexity or smoothness of f is used for the reduction itself.

4. Entropy theory for the Burgers flux

Rankine–Hugoniot alone does not select physical shocks: the step admits weak solutions in both compressive and expansive configurations. The entropy condition removes the non-physical ones. We formalise the Lax theory for the Burgers flux $f(u) = u^2/2$ with the standard convex entropy pair.

```
noncomputable def burgersFlux (u : ℝ) : ℝ := u ^ 2 / 2
noncomputable def entropy (u : ℝ) : ℝ := u ^ 2 / 2
noncomputable def entropyFlux (u : ℝ) : ℝ := u ^ 3 / 6

noncomputable def entropyDissipation (uL uR s : ℝ) : ℝ :=
s * (entropy uR - entropy uL) + entropyFlux uL - entropyFlux uR
```

At the Rankine–Hugoniot speed the dissipation has a closed form whose sign is governed by the ordering of the states, which is exactly the Lax condition. The package makes the resulting non-uniqueness explicit: the midpoint expansion step is a genuine weak solution of the Burgers equation that nonetheless violates the Lax condition and dissipates entropy of the wrong sign.

```
theorem ConservationLaw.expansion_midpoint_is_weak_but_not_entropic
(T uL uR : ℝ) (h : uL < uR) : ...
```


This is the formal counterpart of the standard textbook observation that weak solutions of conservation laws are not unique and that an admissibility criterion is indispensable. Keeping this pathology inside the verified corpus — rather than in prose — fixes precisely what the entropy condition is needed for.

5. Gradient blow-up for the inviscid Burgers equation

5.1 Regular solutions and the compressive datum

A regular solution is a C^2 function satisfying $\partial_t u + u \partial_x u = 0$ pointwise; the initial datum is the compressive ramp.

```
def initialRamp : ℝ → ℝ := fun x => -x
```

5.2 Characteristics, the Riccati gradient, and blow-up

Along each characteristic $t \mapsto x_0(1 - t)$ the solution is constant, proved through a Lyapunov-energy uniqueness lemma for the underlying linear ODE (constant_along_characteristic). Differentiating the equation in space, the gradient $V(t) = u_x(t, x_0(1 - t))$ satisfies Riccati equation with a compressive initial value, whose exact solution is explicit.

```
-- V' = -V^2 with V 0 = -1
theorem Burgers.gradient_riccati_evolution ...
-- V t = -1 / (1 - t)
theorem Burgers.gradient_eq_neg_one_div ...
```

The exact formula diverges as $t \rightarrow 1^-$, but the final argument avoids limits entirely: given any putative gradient bound M , at the critical time $t = 1 - 1/(|M| + 1)$ the formula yields a gradient of magnitude $|M| + 1 > M$, a direct algebraic contradiction. Hence no regular solution reaches time 1.

```
theorem Burgers.not_isRegularSolution_initialRamp {T : ℝ} (hT : 1 ≤ T)
(u : ℝ → ℝ → ℝ) : ¬ IsRegularSolution initialRamp T u
```

This is, to our knowledge, the first machine-checked theorem of finite-time singularity formation for a nonlinear evolution equation. It is the exact counterpoint to elliptic regularity: where De Giorgi–Nash–Moser produces smoothness, the method of characteristics here produces its destruction, and the mechanism — a Riccati blow-up of the gradient — is displayed in closed form.

6. The KdV soliton and conservation laws

For the normalisation $u_t + u u_x + u_{xxx} = 0$, the travelling-wave ansatz $u(t, x) = f(x - c t)$ reduces the PDE to the stationary soliton ODE $-c f' + f f' + f''' = 0$ (travellingWave_redu). The package develops the hyperbolic-secant calculus (sech, the identity $\text{sech}^2 = 1 - \tanh^2$, and the derivatives of the relevant composites) and verifies the exact profile.

```
theorem KdV.soliton_satisfies_kdv (c : ℝ) (hc : 0 < c) (ξ : ℝ) :
-c * deriv (fun x => 3 * c * sech (Real.sqrt c / 2 * x) ^ 2) ξ +
(3 * c * sech (Real.sqrt c / 2 * ξ) ^ 2) *
deriv (fun x => 3 * c * sech (Real.sqrt c / 2 * x) ^ 2) ξ +
deriv (fun y => deriv (fun z =>
deriv (fun x => 3 * c * sech (Real.sqrt c / 2 * x) ^ 2) z) y) ξ = 0
```

For compactly supported smooth solutions the package proves the rate forms of the first two conservation laws — mass and quadratic energy —

```
theorem KdV.ConservedSolution.massRate_conserved -- ∫ u_t = 0
theorem KdV.ConservedSolution.energyRate_conserved -- ∫ u u_t = 0
```

obtained from the integral identities $\int u^k u_x = 0$ and the double integration by parts $\int u u_{xxx} = 0$ for compactly supported integrands. The soliton and the conservation laws together give the smallest non-trivial verified fragment of integrable-systems theory.

7. Formalisation notes

Generality of the flux. The weak-solution API and the shock reduction are stated for an arbitrary $f : \mathbb{R} \rightarrow \mathbb{R}$. Convexity enters only in Section 4, and only to obtain the sign of the entropy dissipation; the Burgers-specific results live in a single module (`Burgers.lean`) and never contaminate the general theory.

Measure-theoretic choices. The residual integrates over the open cylinder with the ambient volume on $\mathbb{R} \times \mathbb{R}$; test functions have compact support interior to the cylinder, so no boundary terms arise and all integrals are over bounded sets after restricting to the support. Local integrability of the solution and of its flux is part of the definition of a weak solution.

No axioms, no `sorry`, no `native\_decide`. Every result is kernel-checked. The numerical witnesses in the Burgers and KdV packages are proved by `norm_num` and explicit finite reductions rather than by trusted compiled evaluation, so the axiom certificate of each headline theorem is exactly `[propext, Classical.choice, Quot.sound]`.

Independence and PR partition. The three packages share no code and depend only on Mathlib; the four calculus helpers common to `ConservationLaws` and `BurgersBlowUp` are duplicated deliberately so that each package is self-contained, and would be contributed once, in their most general form, at PR time. Within each package the modules form a linear chain, and the intended Mathlib PR series follows that chain, each PR self-contained and useful on its own (definitions and the null solution; test-function calculus; the Galilean frame and Fubini cancellations; the shock profile and Rankine–Hugoniot; the exact reduction; the entropy theory). The KdV special-function lemmas (`sect` and its derivatives) are split off first, as they belong in `Analysis/SpecialFunctions` independently of the PDE.

8. Related work

The formalisation of De Giorgi–Nash–Moser (2026) established elliptic regularity in Lean and is the closest precedent for non-trivial PDE in a proof assistant; the present work is complementary, addressing evolutionary hyperbolic and dispersive equations. Mathlib already provides the ODE uniqueness and Grönwall inequalities used in Section 5, the smoothness and compact-support API used throughout, and the measure theory underlying the residual. Classical references for the mathematics formalised here are Lax's theory of hyperbolic conservation laws and the shock admissibility conditions [2], the method of characteristics for Burgers' equation, and the inverse-scattering theory of KdV [4]; we formalise only the elementary, self-contained fragments of each.

9. Availability

The three packages are available at <https://github.com/Alektronnik/M4TH> under the Apache 2.0 license, each with its own `lakefile.toml` (pinned to Mathlib `fabf563a`, Lean 4 `v4.31.0`) and a native Lean-generated SVG cover figure. Each package also includes a `Live` single-file study version. After a successful build the axiom certificate of each headline theorem may be reproduced with

```
#print axioms ConservationLaw.hasShockIntegralReduction
#print axioms Burgers.not_isRegularSolution_initialRamp
#print axioms KdV.soliton_satisfies_kdv
```

each returning `[propext, Classical.choice, Quot.sound]`.

References

1. S. Armstrong et al., *A formalisation of the De Giorgi–Nash–Moser theorem in Lean* (2026).

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3. C. M. Dafermos, *Hyperbolic Conservation Laws in Continuum Physics* , Springer,

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4. G. B. Whitham, *Linear and Nonlinear Waves* , Wiley, 1974.

5. The mathlib Community, *The Lean Mathematical Library* , CPP 2020;

<<https://github.com/leanprover-community/mathlib4>>.